

LEÇON PUBLIQUE

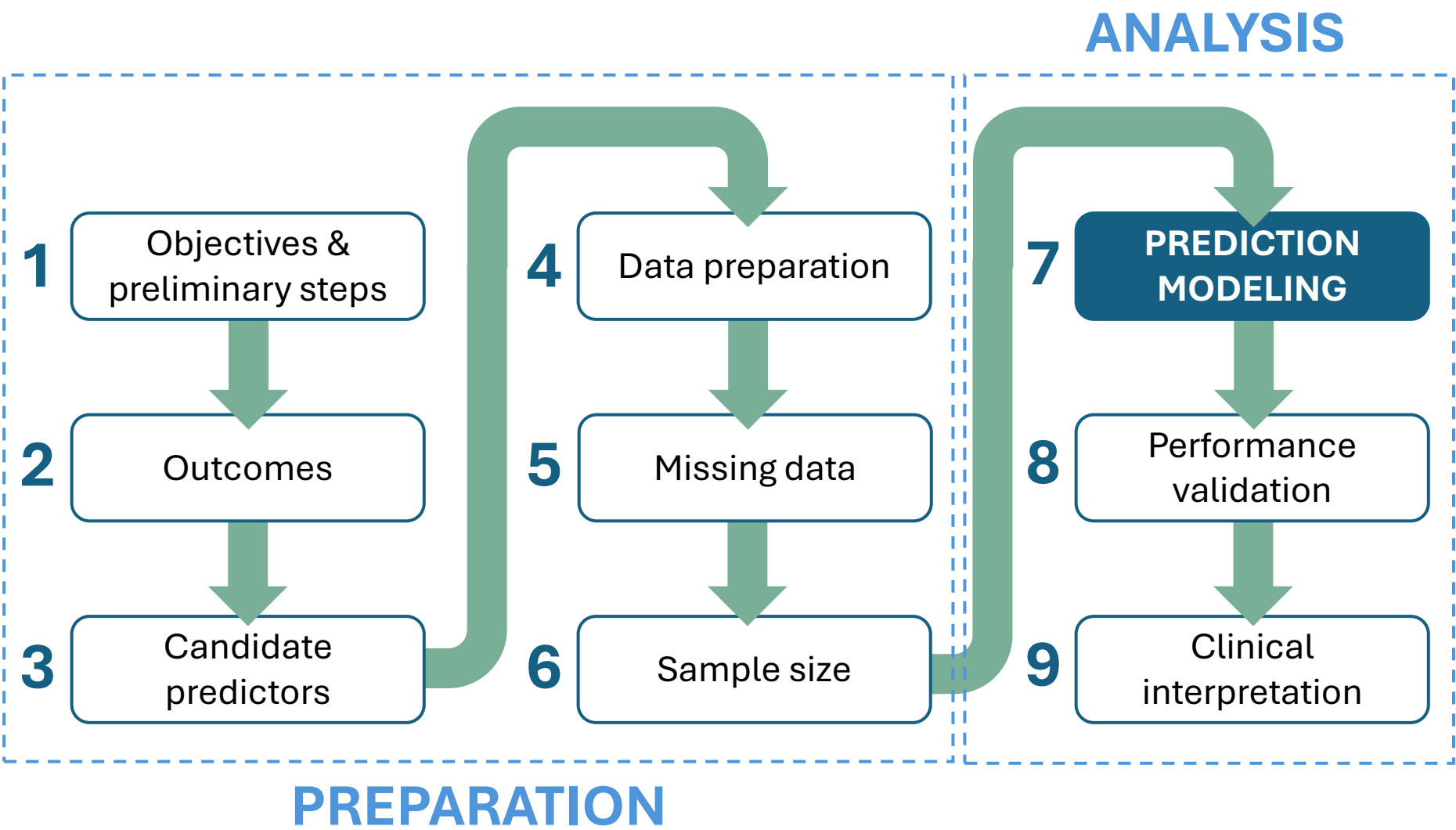
Design, validate, and interpret a predictive model using health data

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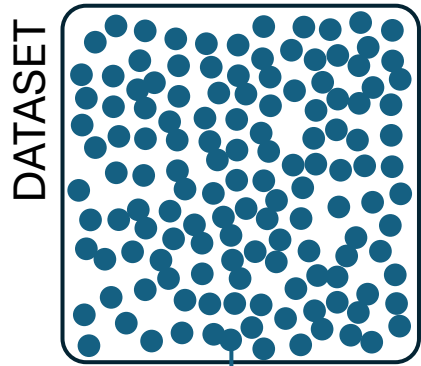
FACULTÉ DE MÉDECINE
UNIVERSITÉ DE NAMUR

Audition pour le poste académique en Analytics and Artificial Intelligence in Health (0.6 ETP)

7. PREDICTION MODELING



7. PREDICTION MODELING : General principle



1. Model the relationship between predictors and the outcome.

$$z = \beta_0 + \beta_1 \cdot X_1 + \beta_2 \cdot X_2 + \beta_3 \cdot X_3 + \dots$$

Where: z is the linear predictor/logit provided by the model
 β are the coefficients associated with the predictors
 X are the predictors

FOR A SPECIFIC
INDIVIDUAL



2. You will use the values of the predictor to calculate the z value, and then, predict the outcome.
 - For **CONTINUOUS outcome**: z = the predicted value of the outcome
 - For **BINARY outcome**, you would like to predict the PROBABILITY that the outcome will occur.



z is not a probability!!! It is the logit: $\log\left(\frac{p}{1-p}\right)$.

$$>> P(Y = 1) = \frac{1}{1+e^{-z}}$$

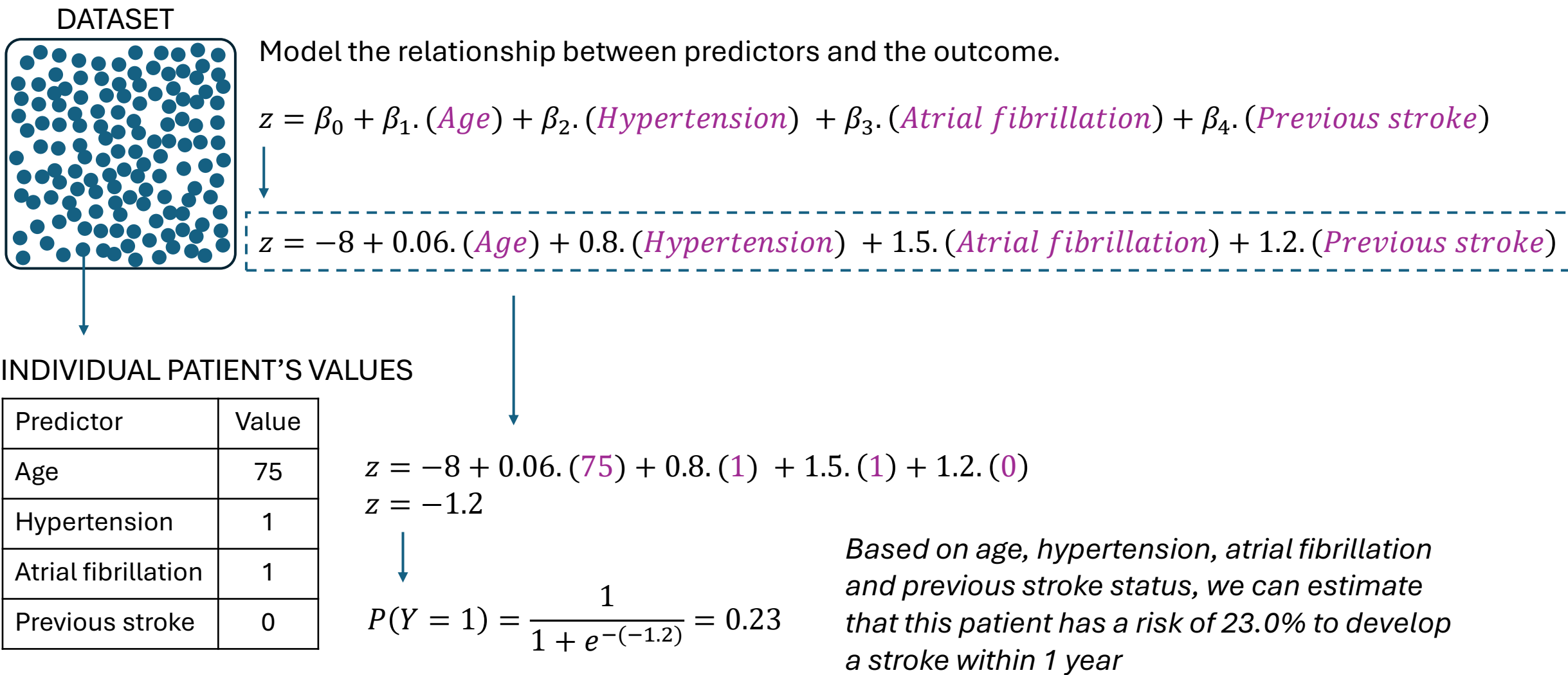
Where: $P(Y = 1)$ is the probability that the outcome (Y) occurs (Y=1)

- For **time-dependent outcome**, you would like to predict the HR.

$$>> HR = e^z$$

7. PREDICTION MODELING : General principle

Objective: Predicting 1-year stroke risk (binary outcome)



7. PREDICTION MODELING : General principle

How to build your model in practice? 4 Steps:

- A. Choose the model type
- B. Select candidate predictors
- C. Address overfitting and underfitting
- D. Estimate the model coefficients (penalisation/shrinkage)

7. PREDICTION MODELING : A. Select model

A. Select the methods according to your outcome. BASICALLY:

Continuous outcomes
e.g., systolic blood pressure



LINEAR REGRESSION MODEL



R Code

```
</> R  
  
model <- lm(  
  outcome ~ pred1 + pred2 + pred3  
    + pred4,  
  data = data  
)
```

Binary outcomes
e.g., stroke (yes/no)



LOGISTIC REGRESSION MODEL



R Code

```
</> R  
  
model <- glm(  
  outcome ~ pred1 + pred2 + pred3  
    + pred4,  
  family = binomial(),  
  data = data  
)
```

Survival outcomes
e.g., time to hospitalisation



COX REGRESSION MODEL



R Code

```
</> R  
  
model <- coxph(  
  Surv(time, outcome) ~ pred1 +  
    pred2 + pred3 + pred4,  
  data = data  
)
```

7. PREDICTION MODELING : A. Select model

> **OTHER OPTIONS CAN BE USED** (according to your outcome...)

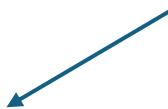
- Poisson regression
- Mixed-effects models
- ...



7	REGRESSION ANALYSES
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> Inclusion of **NON-LINEAR TERMS** for continuous outcomes, or **INTERACTIONS** between predictors.
If dataset has a sufficient sample size!

Example: $z = \beta_0 + \beta_1 \cdot X_1 + \beta_2 \cdot X_2 + \beta_3 \cdot (X_1 \times X_2)$



> **MACHINE LEARNING MODELS?**

Yes! (e.g., random forests, support vector machines, boosting methods, neural network)

But: more advanced, require huge datasets, not « immune to biases »



9	PREDICTION (2)
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7. PREDICTION MODELING : B. Select predictors

B. Select the predictors in your model.

You have already selected « **CANDIDATE PREDICTORS** » (your list of prespecified predictors from step 3)

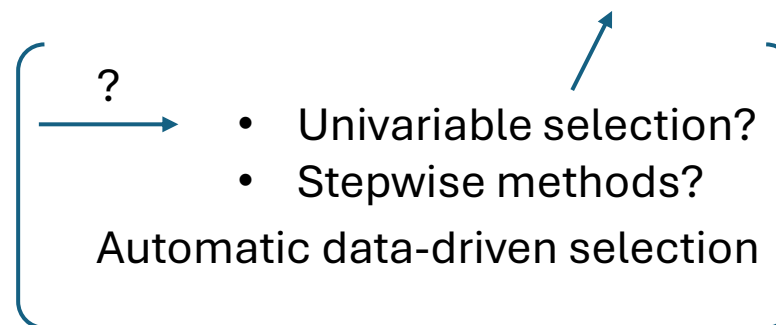
- Causal or non-causal predictors
- Importance of clinical reasoning
- Importance of implementation thinking!



Should we include all candidate predictors in the model?



Yes!



- Loss of information
- Estimation bias
- Worse predictive performance



To be avoided!

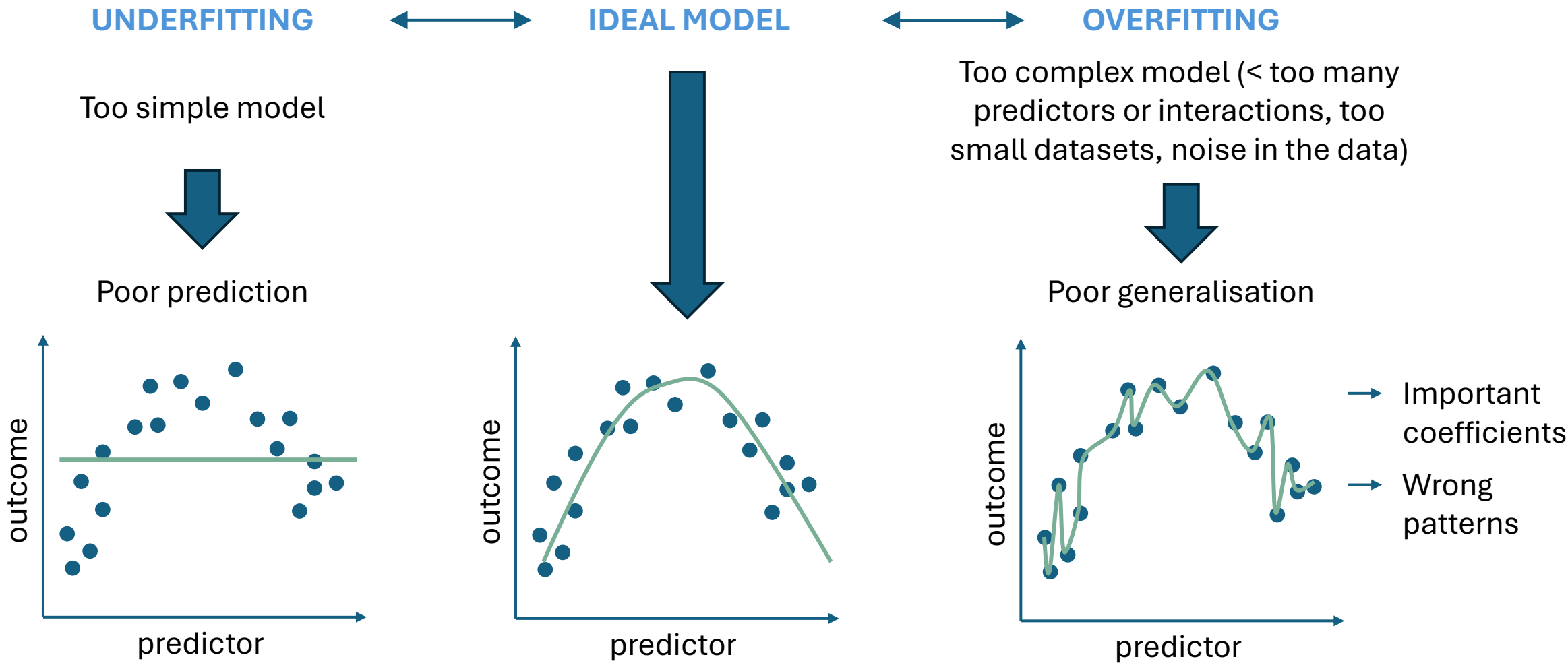
No real limitation in number of predictors included, but model complexity must remain compatible with sample size!



OVERFITTING !!

7. PREDICTION MODELING : C. Overfitting vs Underfitting

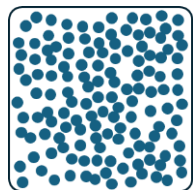
The challenge of model development is to find the right **balance** between simplicity and complexity.



7. PREDICTION MODELING : D. Model estimation

D. Model estimation = estimate the coefficients of your model!

Example: Model the relationship between predictors (Age, Hypertension, Atrial fibrillation, and previous stroke) and having a Stroke within 1 year.



→
$$z = \beta_0 + \beta_1 \cdot (\text{Age}) + \beta_2 \cdot (\text{Hypertension}) + \beta_3 \cdot (\text{Atrial fibrillation}) + \beta_4 \cdot (\text{Previous stroke})$$

R Code

```
</> R
fit <- glm(
  out_stroke ~ age + hypertension + af
  + prev_stroke,
  family = binomial(),
  data = data
)
summary(fit)
```

R Output

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-8.00	2.10	-3.81	<0.001
age	0.06	0.02	3.00	0.003
hypertension	0.80	0.50	1.60	0.110
af	1.50	0.55	2.73	0.006
prev_stroke	1.20	0.60	2.00	0.046

For Logistic models: Maximum Likelihood Estimation (MLE)

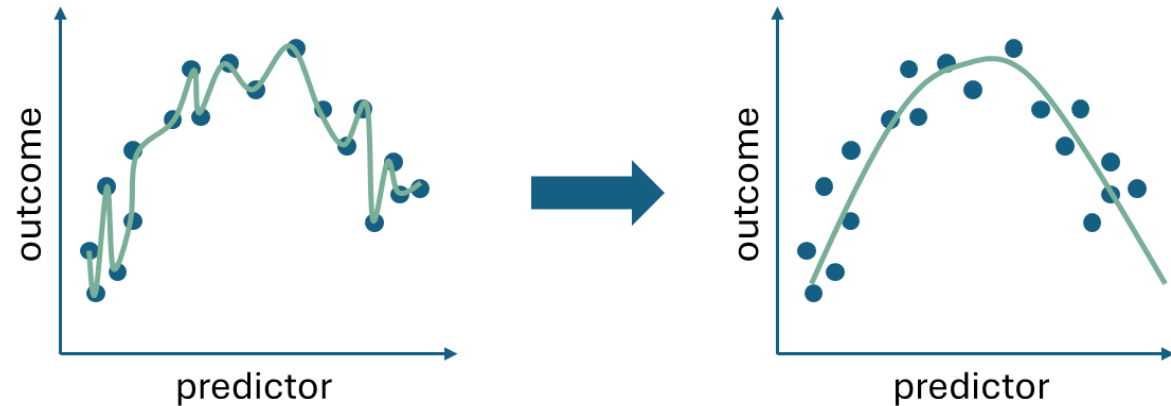
7. PREDICTION MODELING : D. Model estimation



OVERFITTING !!

The solution to avoid overfitting? → **PENALISATION** (or regularisation, shrinkage)

Penalisation **discourages large coefficients** in order to reduce overfitting and improve generalization to new data.



The result :

- Lead to smaller absolute values of the coefficients
- Exclusion of some predictors (if coefficients is set to 0!) – *for LASSO or Elastic Net*
- It will NOT change the model equation, it changes HOW the coefficient are estimated.

>>> **Simpler models, more interpretable!**

Different methods :

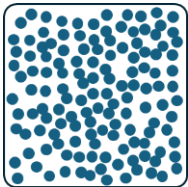
- Ridge
- **LASSO**
- Elastic Net

Shrinks coefficients
Shrinks + variable selection
Combination of both

It can be applied to all methods
(Linear, Logistic, Cox,...).

7. PREDICTION MODELING : D. Model estimation

In practice? **Example:** Model the relationship between predictors (Age, Hypertension, Atrial fibrillation, and previous stroke) and having a Stroke within 1 year. (**PENALISED MODEL: LASSO**)



→
$$z = \beta_0 + \beta_1 \cdot (\text{Age}) + \beta_2 \cdot (\text{Hypertension}) + \beta_3 \cdot (\text{Atrial fibrillation}) + \beta_4 \cdot (\text{Previous stroke})$$

R Code

```
</> R

# X = predictor matrix
X <- model.matrix(
  out_stroke ~ age + hypertension + af + prev_stroke,
  data = data
)[, -1]

# Y = outcome
Y <- data$out_stroke

fit_lasso <- cv.glmnet(
  x ~ X,
  y = Y,
  family = "binomial",
  alpha = 1
)

coef(fit_lasso, s = "lambda.min")
```

→ λ = intensity of the penalisation
 $\lambda = 0$: no penalisation; $\lambda = \infty$: max penalisation
« lambda.min » vs. « lambda.1se »

R Output

Coefficients:

(Intercept)
age
hypertension
af
prev_stroke

Estimate
-6.80
0.05
.
1.20
0.90

Coefficients are shrinked

One coefficient has been excluded

No p-value?

7. PREDICTION MODELING: Significance of the model?

Should we speak of statistical significance? **Where are the p-values?**

We are not in explanatory research (we are not testing a hypothesis)



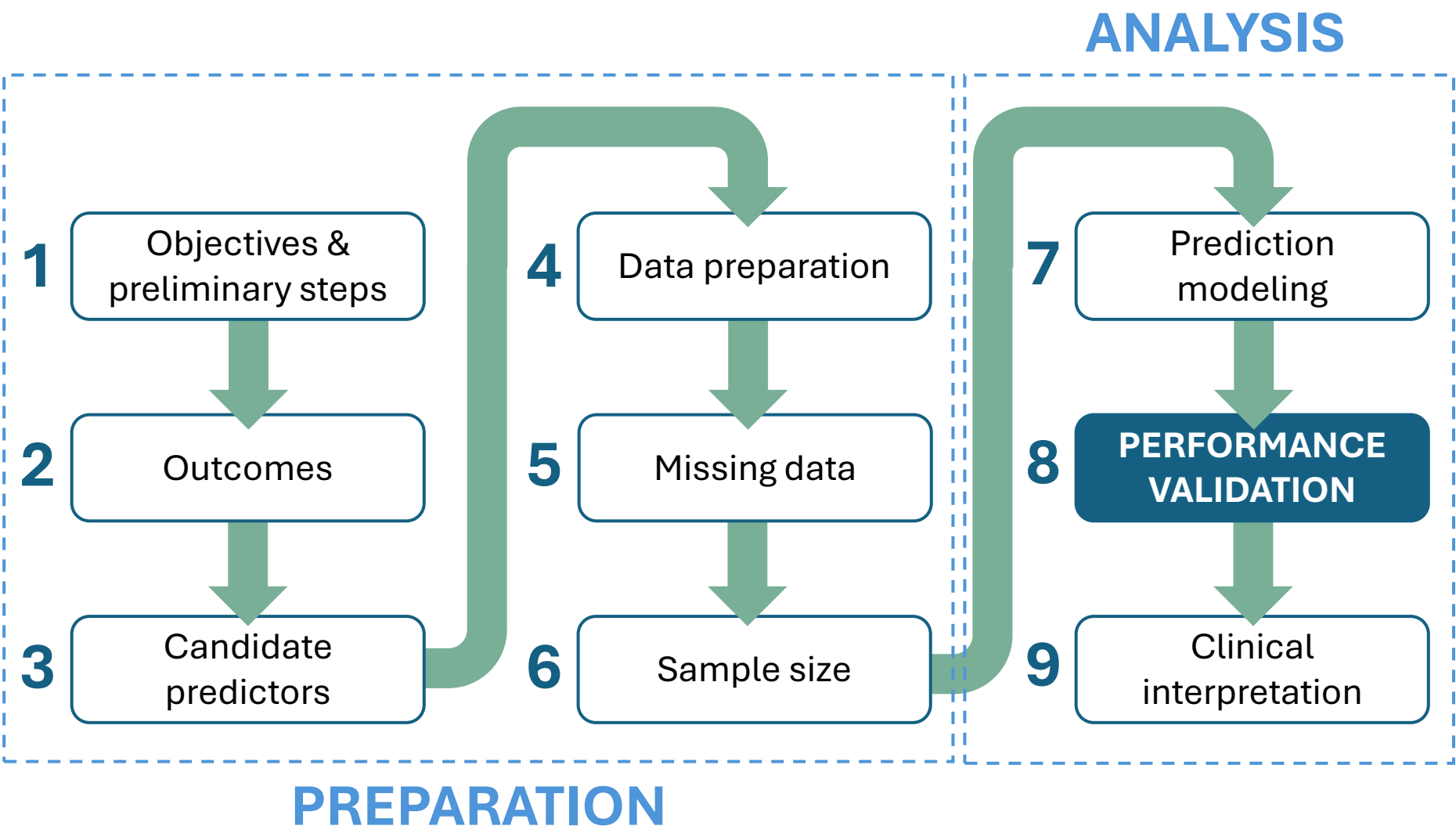
We will not assess significance!

The goal is to know if the model accurately predict outcomes in new individuals!



We will assess **PERFORMANCE!**

8. PERFORMANCE VALIDATION

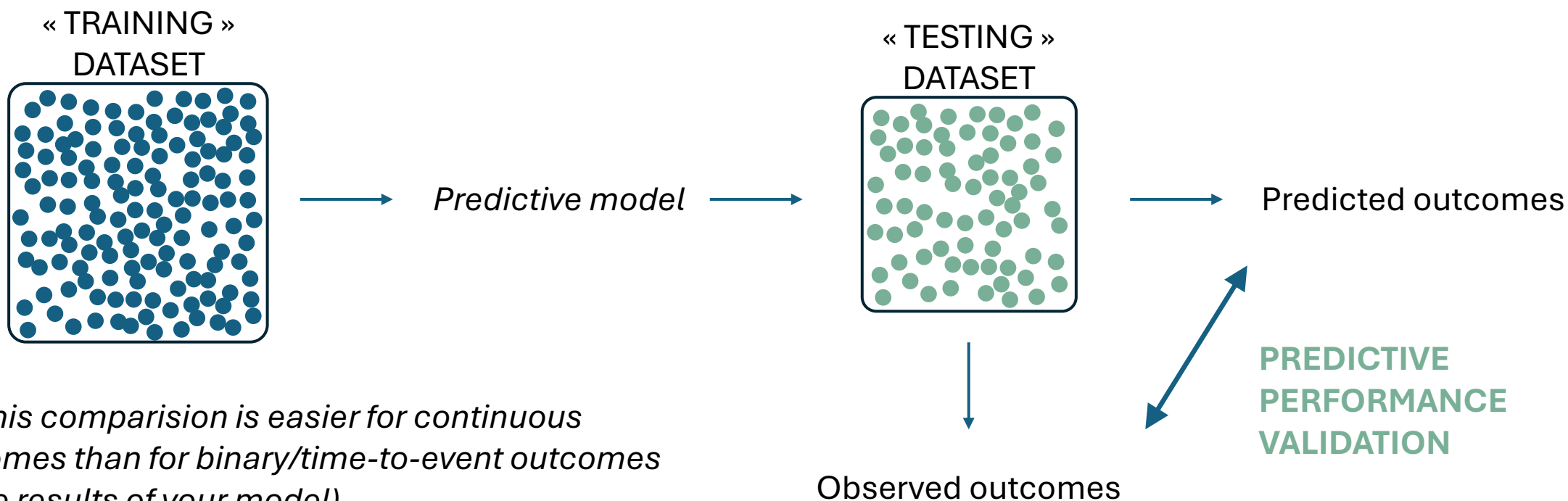


8. PERFORMANCE VALIDATION : General principle

We would like to know if the model is predicting correctly the outcome.



VALIDATION = The process of evaluation of the **PREDICTIVE PERFORMANCE** of the model



NB: this comparison is easier for continuous outcomes than for binary/time-to-event outcomes (cf the results of your model)

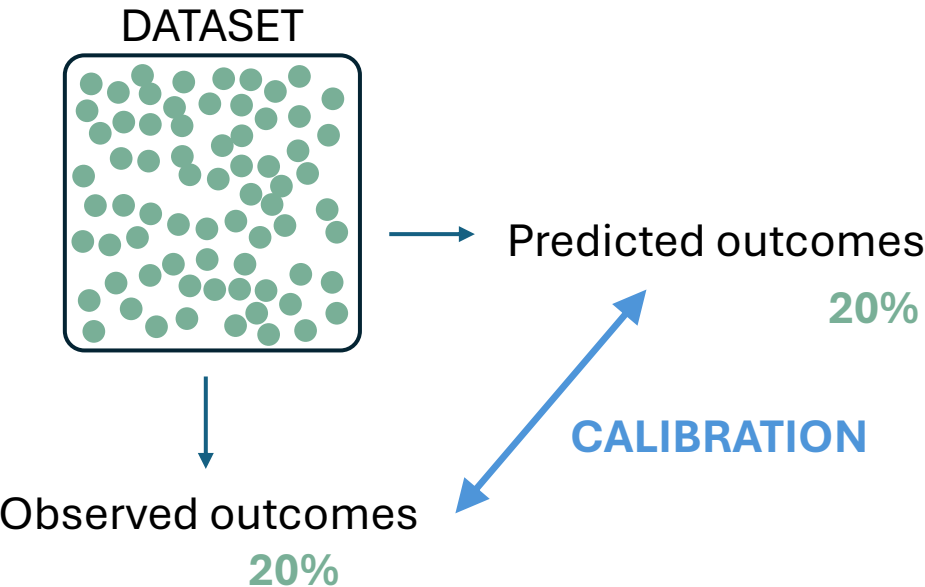
8. PERFORMANCE VALIDATION : A. Measuring performance / Calibration vs. Discrimination

Two different dimensions of prediction performance (both should be assessed!):

1. CALIBRATION

The agreement between observed and predicted outcome values.

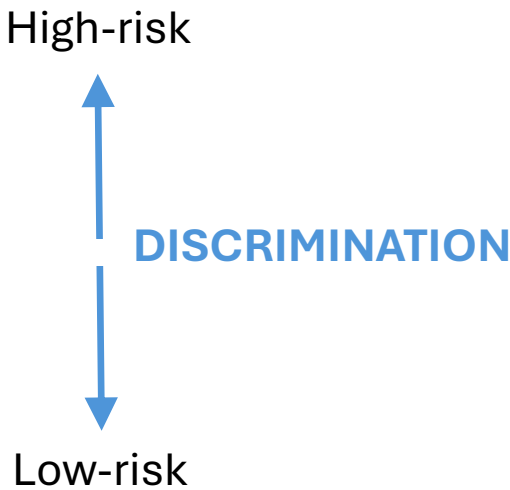
>> *Are predicted risks numerically correct?*



2. DISCRIMINATION

The model's ability to distinguish between patients with different outcomes.

>> *Can the model separate high-risk from low-risk people?*



8. PERFORMANCE VALIDATION : A. Measuring performance / Calibration vs. Discrimination

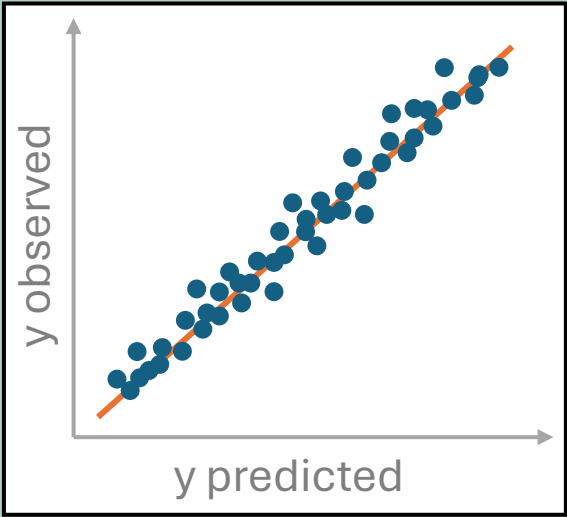
	CONTINUOUS OUTCOMES	BINARY OUTCOMES	TIME-TO-EVENT OUTCOMES
« Good » CALIBRATION	Predicted values do not systematically overestimate or underestimate observed values	The model does not overestimate or underestimate risks	The model does not overestimate or underestimate risks
« Good » DISCRIMINATION	High predicted values in high observed values	Separate people at high risk from those at low risk of event	Rank individuals according to their survival

8. PERFORMANCE VALIDATION : A. Measuring performance / The measures

1. CALIBRATION

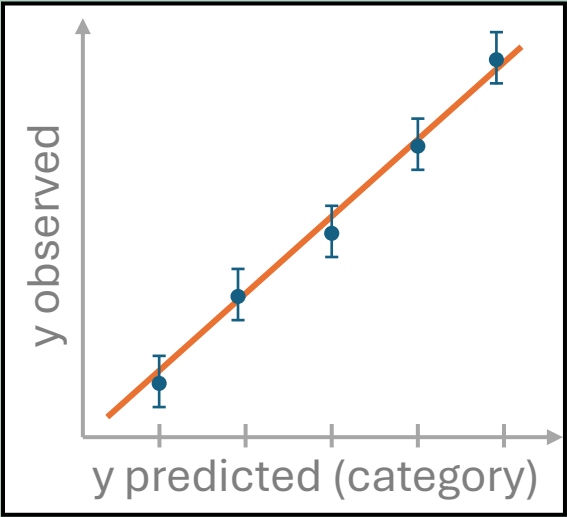
CONTINUOUS OUTCOMES

Scatter plot



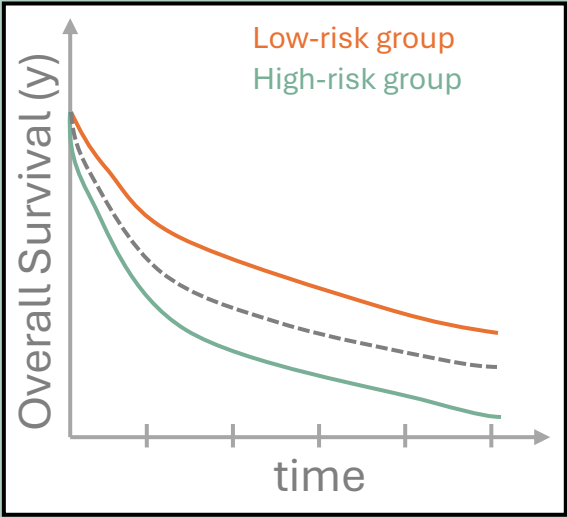
BINARY OUTCOMES

Calibration plot



TIME-TO-EVENT OUTCOMES

Kaplan Meier



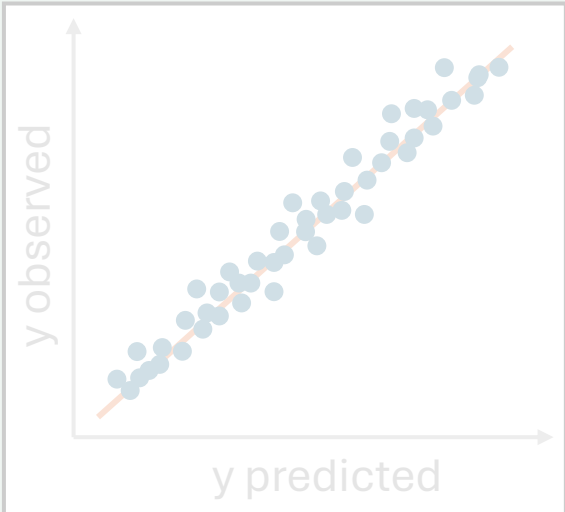
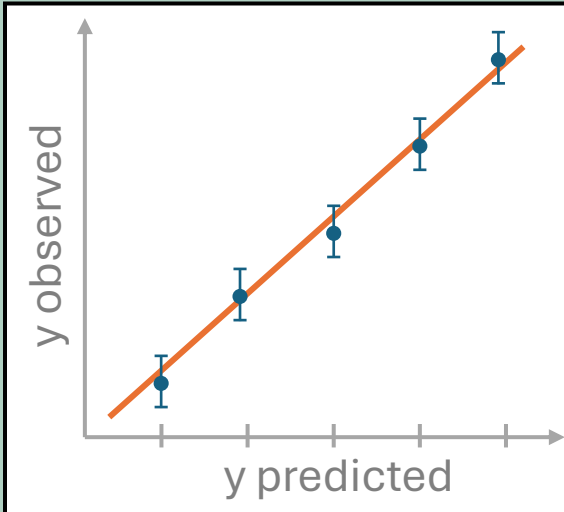
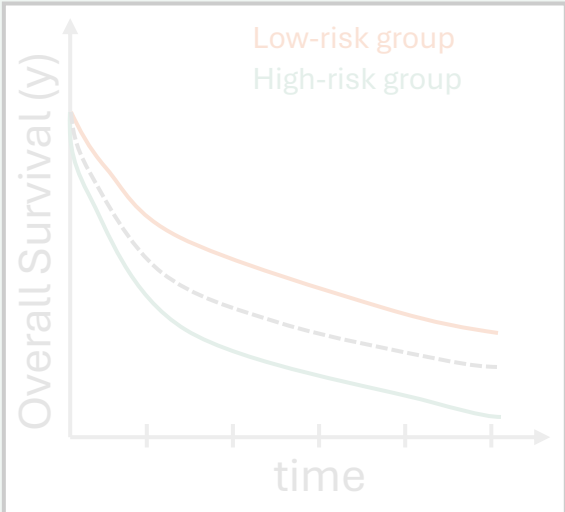
2. DISCRIMINATION

Rank correlation statistics

Area Under Receiver
Operating Curve (AUROC)
or C-statistics

Harrell's C-statistics

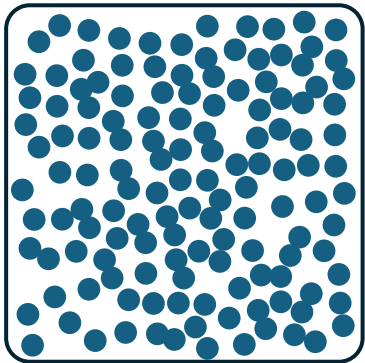
8. PERFORMANCE VALIDATION : A. Measuring performance / The measures

	CONTINUOUS OUTCOMES	BINARY OUTCOMES	TIME-TO-EVENT OUTCOMES
1. CALIBRATION	Scatter plot 	Calibration plot 	Kaplan Meier 
2. DISCRIMINATION	Rank correlation statistics	Area Under Receiver Operating Curve (AUROC) or C-statistics	Harrell's C-statistics

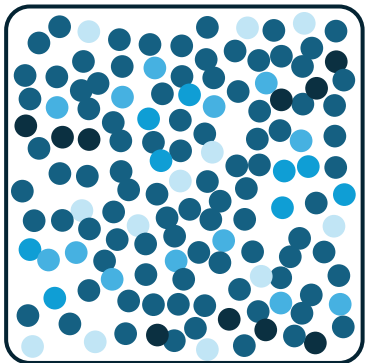
8. PERFORMANCE VALIDATION : A. Measuring performance / Calibration of binary outcome

1. CALIBRATION : Calibration plot

- A. Stratify your dataset according to the prediction.
- B. Within each strata, report the proportion of individuals with the observed outcome.
- C. Build your plot.



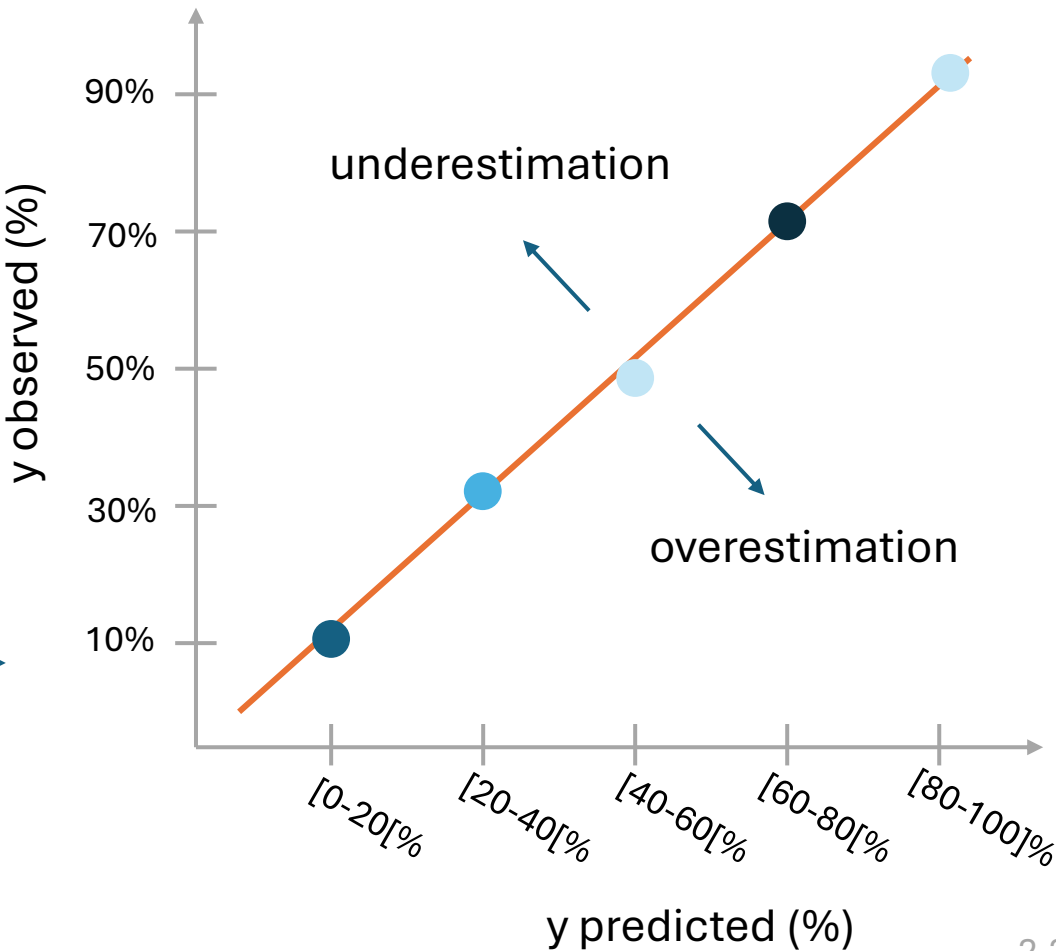
A.



B.

y predicted (%)	y observed (%)
● [0-20[%	10.2%
● [20-40[%	31.0%
● [40-60[%	49.5%
● [60-80[%	70.4%
● [80-100] %	91.0%

C.



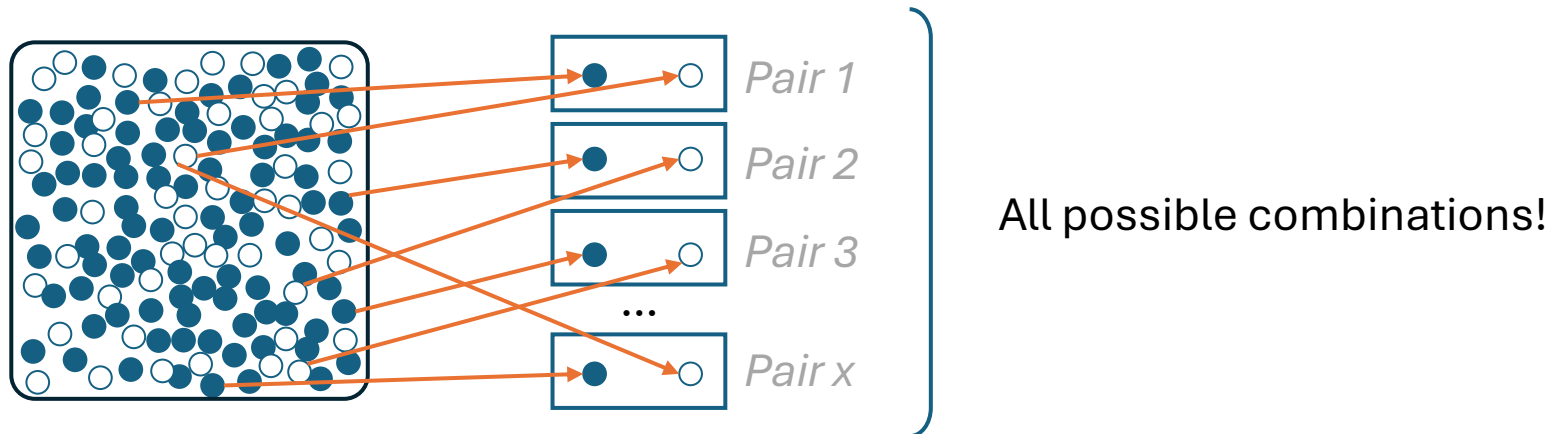
8. PERFORMANCE VALIDATION : A. Measuring performance / Discrimination of binary outcome

2. DISCRIMINATION : Area Under Receiver Operating Curve (AUROC)

= The probability that a randomly selected patient with the event has a higher predicted risk than a randomly selected patient without the event

For our example (predicting the risk of stroke within 1 year), the question is: « *Do patients who experience stroke receive higher predicted risks than patients who do not?* »

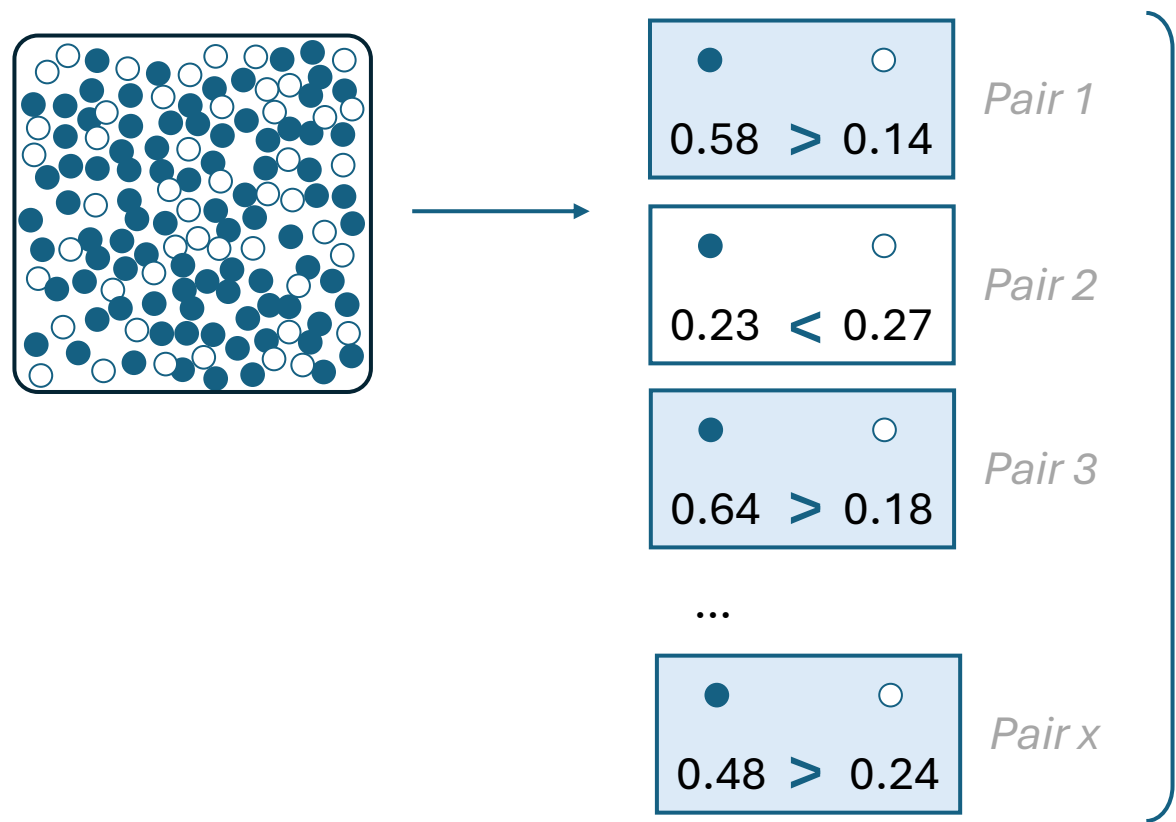
The principle: **A.** we will form pair of patients from the dataset according to the outcome observed (1 with [Y=1] ● + 1 with [Y=0] ○).



8. PERFORMANCE VALIDATION : A. Measuring performance / Discrimination of binary outcome

2. DISCRIMINATION : Area Under Receiver Operating Curve (AUROC)

- B. For each pair, we will compare the PREDICTED value between patients with Y=1 and Y=0
- C. We will compute the proportion of pairs with predicted value HIGHER in patients with Y=1 than in patients with Y=0. This proportion is the AUROC value [0;1].



Example: AUROC = 0.8

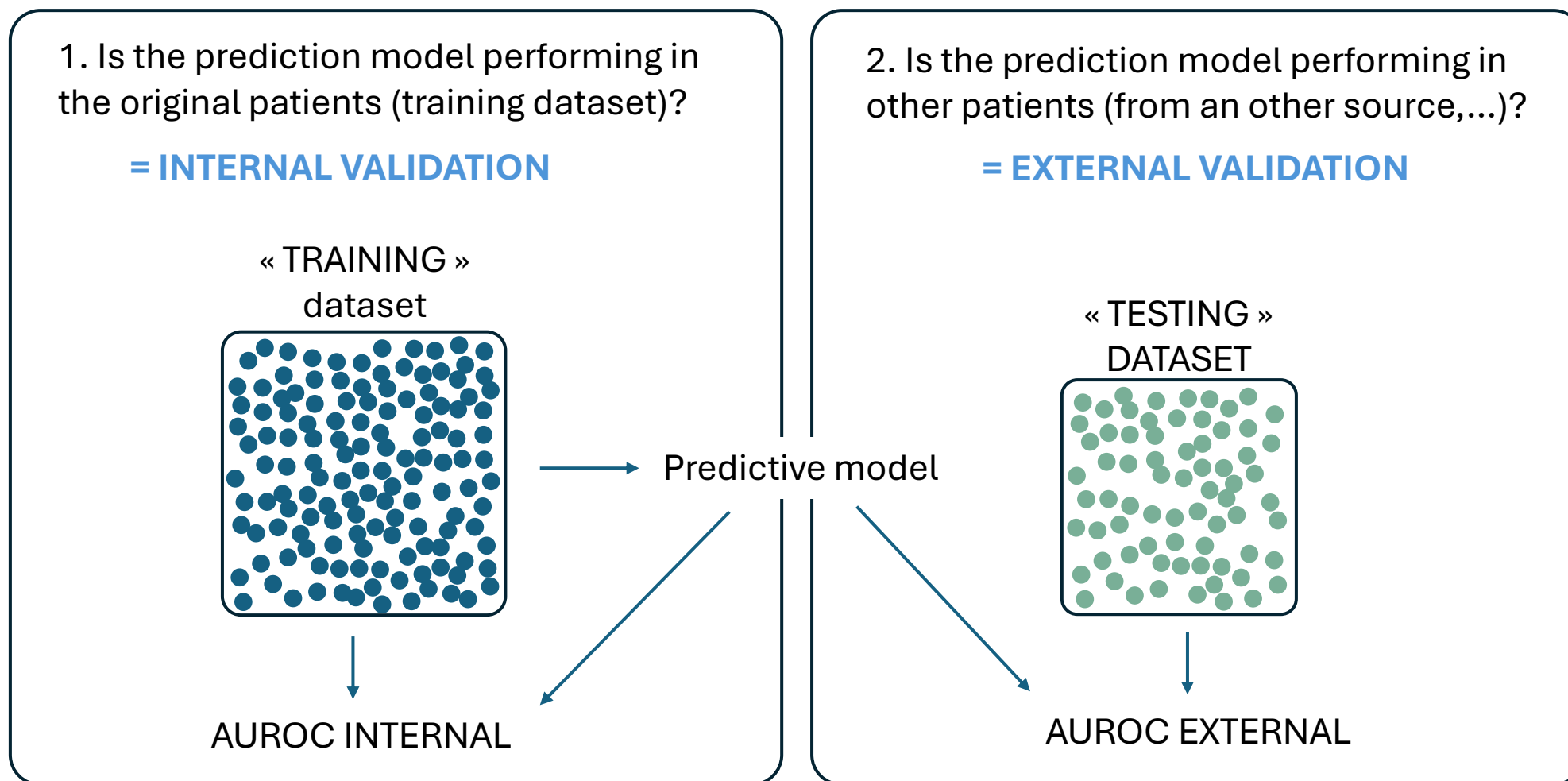
= 80% of patients were correctly classified (i.e., had a higher predicted risk when they had the outcomes, and a lower risk when they had not the outcome)

AUROC interpretation

AUROC	Meaning
0.50 – 0.69	Random guessing
0.70 – 0.79	Acceptable discrimination
0.80 – 0.89	Good discrimination
0.90 – 0.99	Excellent discrimination
1.0	Perfect discrimination

8. PERFORMANCE VALIDATION : B. Validation methods

There are 2 different things you would like to evaluate:

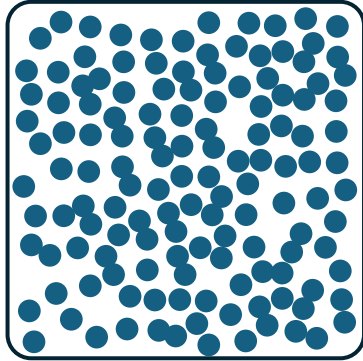


>> The DATA you will use to assess the predictive performance of the model is crucial.

8. PERFORMANCE VALIDATION : B. Validation methods / Internal validation

1. INTERNAL VALIDATION: OPTION 1. SAME DATA FOR TRAINING AND TESTING = *APPARENT VALIDATION*

« TRAINING » and
« TESTING » DATASET



Predictive model



Performance evaluation: AUROC = **0.95 (Excellent discrimination)**

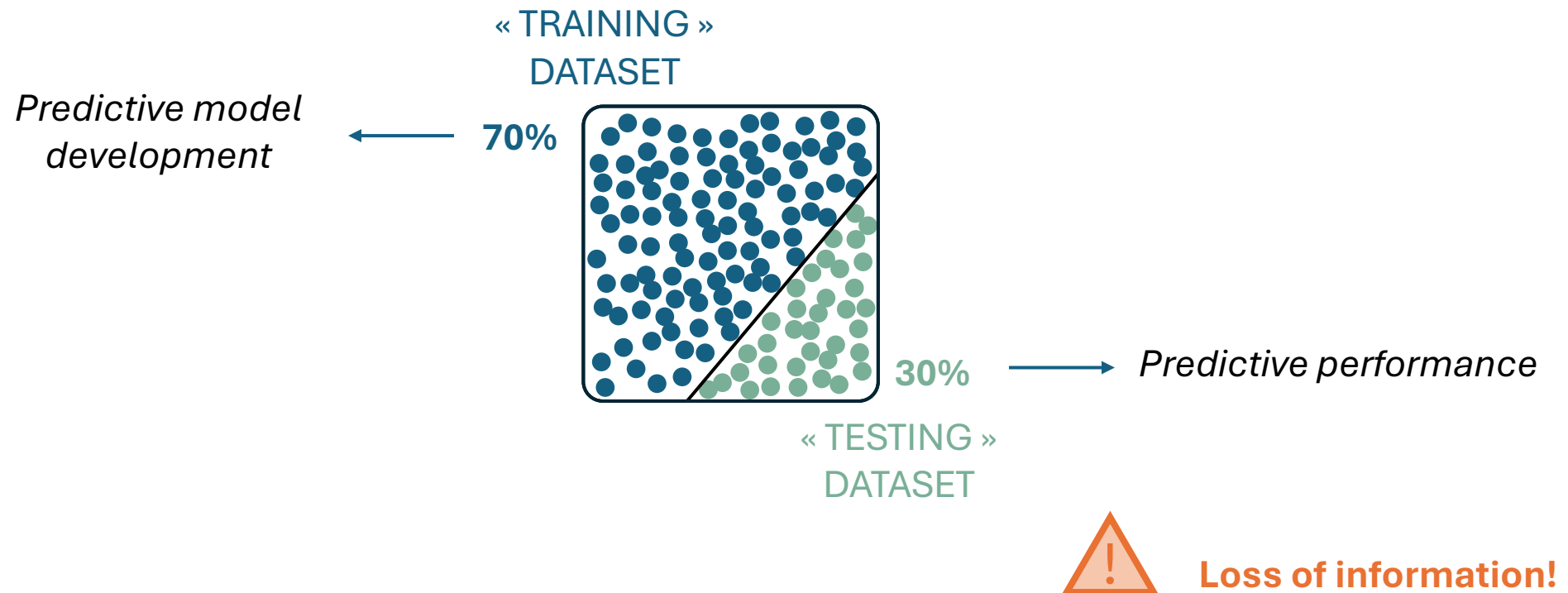


Risk of OVERESTIMATION of the model performance !!

= OPTIMISM

8. PERFORMANCE VALIDATION : B. Validation methods / Internal validation

1. INTERNAL VALIDATION: OPTION 2. SPLIT-SAMPLE



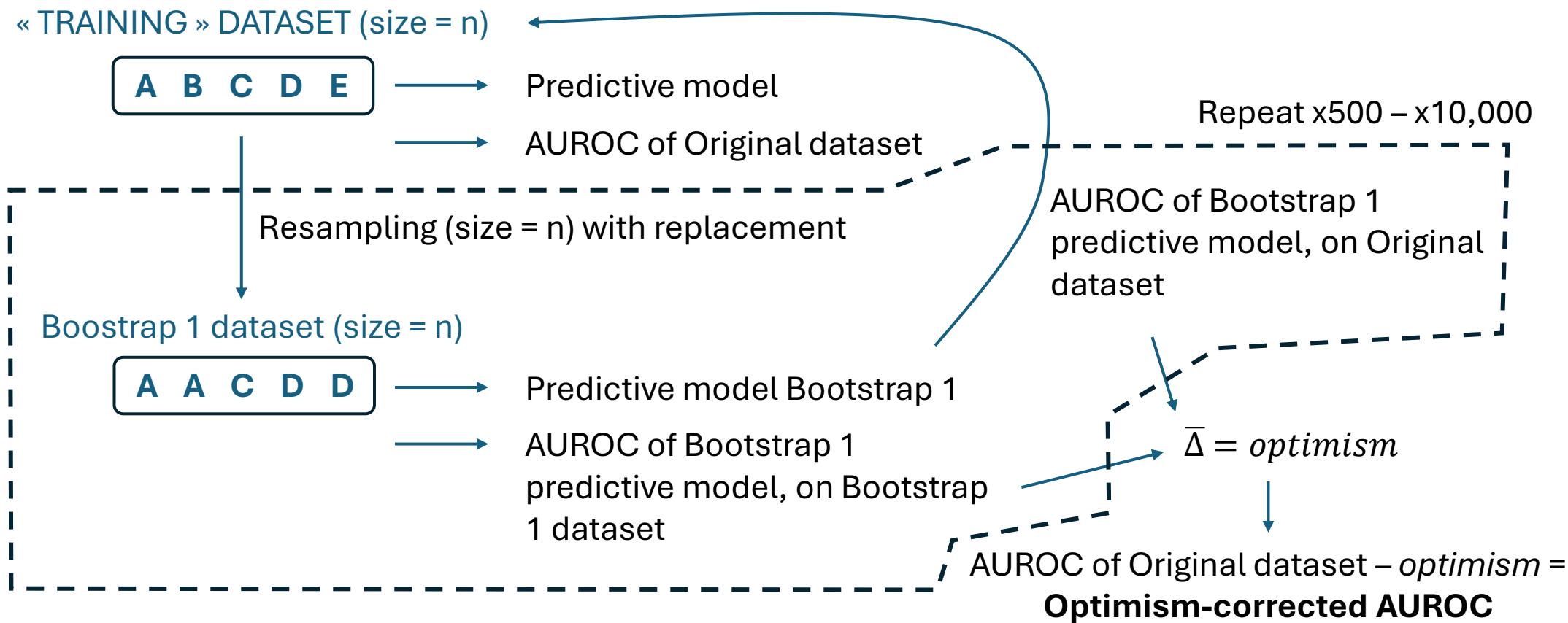
8. PERFORMANCE VALIDATION : B. Validation methods / Internal validation

"to pull oneself up by one's bootstraps"



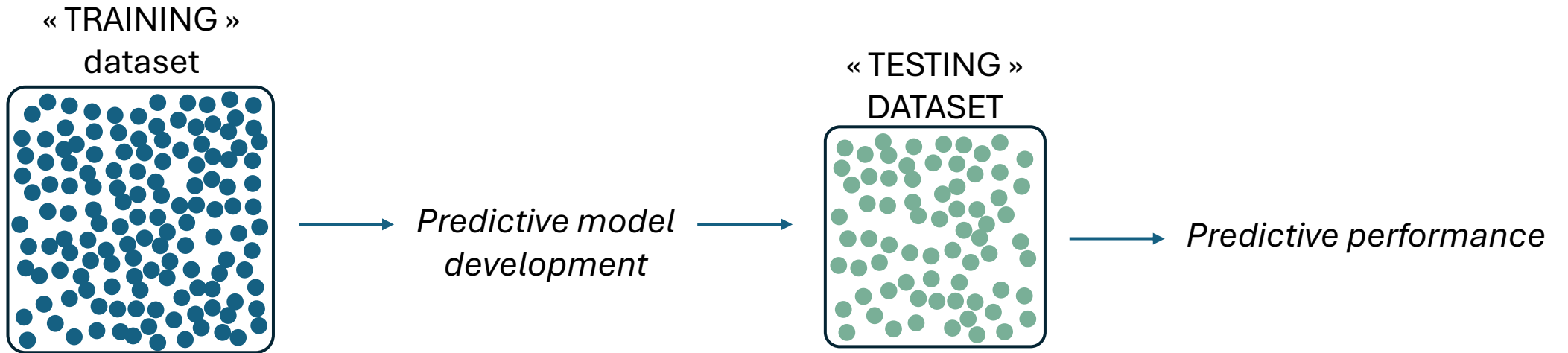
1. INTERNAL VALIDATION: OPTION 3. BOOTSTRAPPING

The goal of the bootstrap process is to quantify the optimism (the « overestimation » of the performance on original data), in order to calculate the bootstrap-corrected AUROC.



8. PERFORMANCE VALIDATION : B. Validation methods / External validation

2. EXTERNAL VALIDATION: Is the model performs in other patients (from a different source, etc)? = GENERALISABILITY of the model.



8. PERFORMANCE VALIDATION : B. Validation methods / Example

Example of reported validation (in a published article)

TABLE 3 | Apparent and validation performance of prediction model.

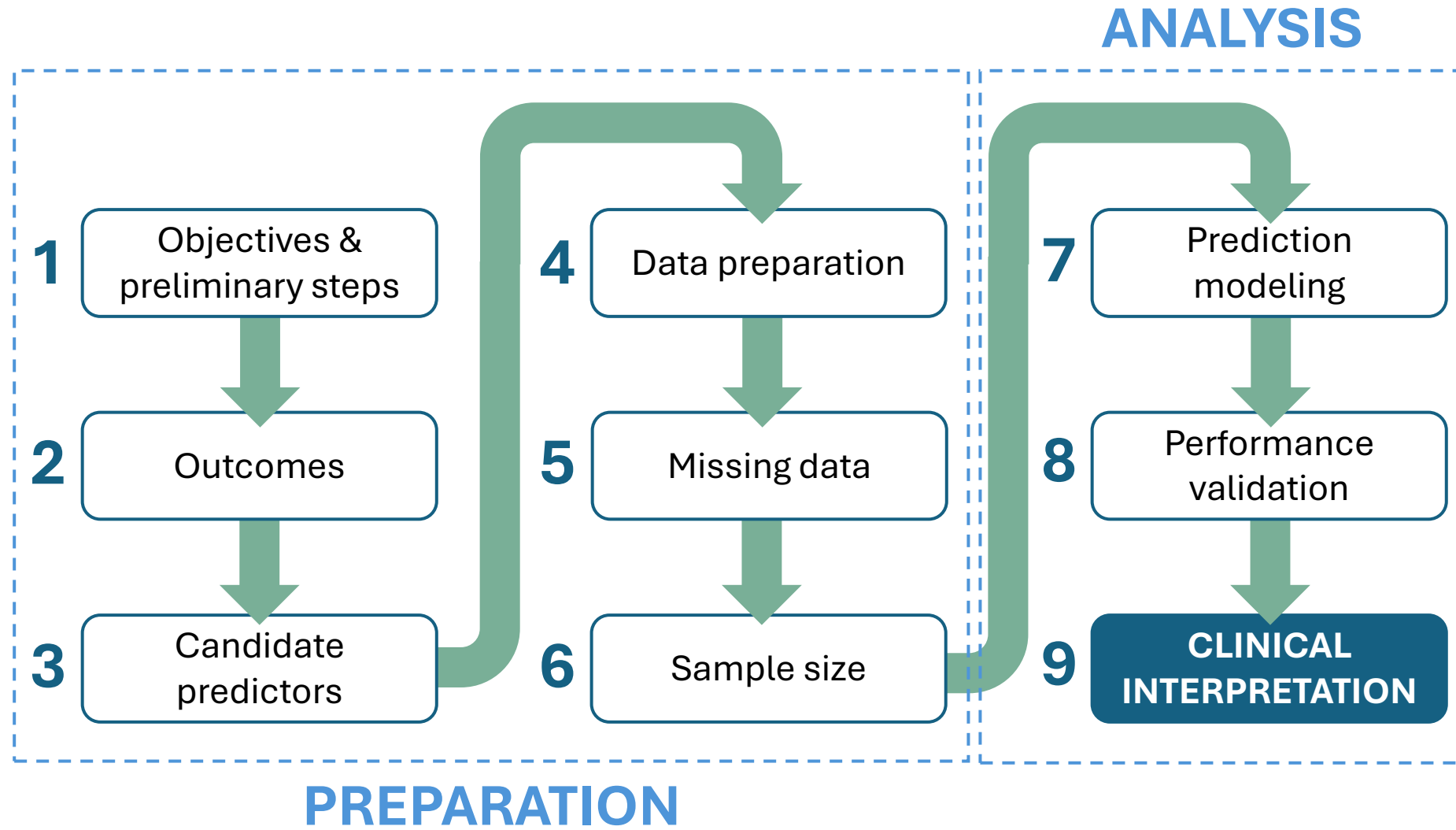
Performance Statistics	Apparent performance in original sample	Internal validation ^a			External validation ^b
		Bootstrap performance	Test performance in original sample	Optimism corrected	
C statistic	0.9147	0.91739	0.91157	0.90887	0.8176
Calibration ^c	Slope: 1.0254 Intercept: 0.0326	Slope: 1.0244 Intercept: 0.0150	Slope: 0.9272 Intercept: −0.1874	Slope: 0.9282	Slope: 0.5986 Intercept: −1.4804

^aPrediction model was internally validated in AECOPD patients admitted between 2015 and 2017. The estimates (95% confidence interval) of apparent performance of prediction model in original sample: 0.9147 (0.8850, 0.9444) for C statistic, 1.0254 (0.7276, 1.3233) for calibration slope and 0.0326 (−0.7834, 0.8486) for intercept. The apparent performance in the model derived from bootstrap sample was compared with the test performance obtained when applying the model to the original sample. Differences in apparent and test performances across all models were averaged to estimate the overall optimism for C statistics and slope. The optimism-corrected C statistics and slope were 0.90887 and 0.9282, respectively. The corrected slope (0.9282) was also shrinkage factor to adjust the original prediction model.

^bExternal validation was performed in AECOPD patients admitted between 2018 and 2019. At external validation, the calibration slope reflects the combined effect of overfitting on the development data (2015–2017) and true differences in effects of predictors. Estimates (95% confidence interval) of prediction model performance were 0.8176 (0.7487, 0.8865) for C statistic, 0.5986 (0.2409, 0.9563) for calibration slope, and −1.4804 (−2.8037, −0.1571) for intercept, respectively.

^cIntercept and slope of calibration plot were estimated in a logistic regression model with in-hospital death events as outcome and linear predictor as the only independent variable.

9. INTERPRETATION



9. INTERPRETATION

> *And now, how do I interpret and use predictions?*

Example:

- You would like to predict the risk of stroke within 1 year.
- You have developed a penalized model, based on predictors (Age, Hypertension, Atrial fibrillation, and previous stroke).

$$z = -5.8 + 0.05 \times (\text{Age}) + 0 \times (\text{Hypertension}) + 1.2 \times (\text{Atrial fibrillation}) + 0.9 \times (\text{Previous stroke})$$

- You have validated your model, internally (Optimism-corrected AUROC = 0.87) and externally (AUROC = 0.83), and you obtained appropriate performance metrics.

Performance statistics	Apparent validation	Internal validation (Optimism-corrected)	External validation
AUROC	0.92	0.87	0.83
Calibration (plot)	Good	Good	Good

9. INTERPRETATION

- You will apply this model to predict the risk of stroke within 1 year in a specific patient, with the following characteristics:

Individual patient characteristics:

Predictor	Value
Age	75
Hypertension	1
Atrial fibrillation	1
Previous stroke	0

$$\text{logit}(P(\text{stroke})) = z = -5.8 + 0.05 \times 75 + 0 + 1.2 \times 1 + 0.9 \times 0 \\ z = -0.85$$



$$P(Y = 1) = \frac{1}{1 + e^{-(-0.85)}} = 0.30$$

- Interpret your model: « *Based on age, hypertension, atrial fibrillation and previous stroke status, we can **estimate** that this patient has a risk of 30% to develop a stroke within 1 year.* »

HANDS-ON SESSION

Hands-on session on Prediction (1)

Duration: 1.5 – 2h

Use the following documents available on WebCampus:

- Support document: ISBMM104_PredMod.pdf
- Dataset 3: ISBMM104_PredMod.csv
- Correction code: ISBMM104_PredMod.R
- R Cheat sheet: ISBMM104_Cheatsheet_PREDICTION.pdf

Answer the following question: “*Develop, validate, and interpret a prediction model to estimate the risk of severe hypoglycaemia in older adults living with type 2 diabetes.*”

SOURCES ET REFERENCES

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- Statistique appliquée à la médecine et à la biologie. Ancelle T. *Statistique appliquée à la médecine et à la biologie*. 5e éd. Paris: Maloine; 2022.
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- How to evaluate a clinical prediction model. Riley RD, Collins GS, van Smeden M, et al. How to evaluate a clinical prediction model. *BMJ*. 2024;385:e074819. doi:10.1136/bmj-2023-074819.